

1605.05775 — formula report

138 inline formulas, 26 display equations, 0 tables, 15 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1605.05775_ EQ0014 (17)	007	■ 0.927 ● W +1	$\begin{aligned} & W_{s_1 s_2 \cdots s_N}^{\ell} = \\ & \sum_{\{\alpha\}} U_{s_1}^{\alpha_1} \cdots U_{\alpha_{i-1} s_i}^{\alpha_i} C_{\alpha_i \alpha_{i+1}}^{\ell} V_{s_{i+1} \alpha_{i+2}}^{\alpha_{i+1}} \cdots V_{s_N}^{\alpha_{N-1}} \end{aligned}$	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} U_{s_1}^{\alpha_1} \cdots U_{\alpha_{i-1} s_i}^{\alpha_i} C_{\alpha_i \alpha_{i+1}}^{\ell} V_{s_{i+1} \alpha_{i+2}}^{\alpha_{i+1}} \cdots V_{s_N}^{\alpha_{N-1}}$	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} U_{s_1}^{\alpha_1} \cdots U_{\alpha_{i-1} s_i}^{\alpha_i} C_{\alpha_i \alpha_{i+1}}^{\ell} V_{s_{i+1} \alpha_{i+2}}^{\alpha_{i+1}} \cdots V_{s_N}^{\alpha_{N-1}}$
1605.05775_ EQ0026 (B4)	010	■ 0.988 ● W +0	$\phi^{\left(s_j\right)}\left(x_j\right)=\sqrt{\left(d-1\right)\left(s_j-1\right)}\left(\cos \left(\frac{\pi}{2} x_j\right)\right)^{d-s_j}\left(\sin \left(\frac{\pi}{2} x_j\right)\right)^{s_j-1}$	$\phi^{s_j}(x_j) = \sqrt{\left(d-1\right)\left(s_j-1\right)}\left(\cos \left(\frac{\pi}{2} x_j\right)\right)^{d-s_j}\left(\sin \left(\frac{\pi}{2} x_j\right)\right)^{s_j-1}$	$\phi^{s_j}(x_j) = \sqrt{\left(d-1\right)\left(s_j-1\right)}\left(\cos \left(\frac{\pi}{2} x_j\right)\right)^{d-s_j}\left(\sin \left(\frac{\pi}{2} x_j\right)\right)^{s_j-1}$
1605.05775_ EQ0013 (15)	007	■ 0.989 ● K +0	$\begin{aligned} & \phi(0)=[1,0] \\ & \phi(1)=[0,1] \end{aligned}$	$\begin{aligned} \phi(0) &= [1, 0] \\ \phi(1) &= [0, 1]. \end{aligned}$	$\begin{aligned} \phi(0) &= [1, 0] \\ \phi(1) &= [0, 1] \end{aligned}$
1605.05775_ EQ0025 (B3)	010	■ 0.990 ● N +0	$\begin{aligned} & \left(\cos^2\left(\theta_j\right)+\sin ^2\left(\theta_j\right)\right)^{d-1}=1 \\ & =\sum_{p=0}^{d-1}\left(\begin{array}{c} d-1 \\ p \end{array}\right)\left(\cos \theta_j\right)^{2(d-1-p)}\left(\sin \theta_j\right)^{2 p} . \end{aligned}$	$\begin{aligned} \left(\cos^2\left(\theta_j\right)+\sin ^2\left(\theta_j\right)\right)^{d-1} &= 1 \\ & =\sum_{p=0}^{d-1}\left(\begin{array}{c} d-1 \\ p \end{array}\right)\left(\cos \theta_j\right)^{2(d-1-p)}\left(\sin \theta_j\right)^{2 p} . \end{aligned}$	$\begin{aligned} \left(\cos^2\left(\theta_j\right)+\sin ^2\left(\theta_j\right)\right)^{d-1} &= 1 \\ & =\sum_{p=0}^{d-1}\left(\begin{array}{c} d-1 \\ p \end{array}\right)\left(\cos \theta_j\right)^{2(d-1-p)}\left(\sin \theta_j\right)^{2 p} . \end{aligned}$
1605.05775_ EQ0005 (5)	003	■ 0.995 ● W +0	$W_{s_1 s_2 \cdots s_N}^{\ell}=\sum_{\{\alpha\}} A_{s_1}^{\alpha_1} A_{s_2}^{\alpha_1 \alpha_2} \cdots A_{s_j}^{\ell ; \alpha_j \alpha_{j+1}} \cdots A_{s_N}^{\alpha_{N-1}}$	$W_{s_1 s_2 \cdots s_N}^{\ell}=\sum_{\{\alpha\}} A_{s_1}^{\alpha_1} A_{s_2}^{\alpha_1 \alpha_2} \cdots A_{s_j}^{\ell ; \alpha_j \alpha_{j+1}} \cdots A_{s_N}^{\alpha_{N-1}}$	$W_{s_1 s_2 \cdots s_N}^{\ell}=\sum_{\{\alpha\}} A_{s_1}^{\alpha_1} A_{s_2}^{\alpha_1 \alpha_2} \cdots A_{s_j}^{\ell ; \alpha_j \alpha_{j+1}} \cdots A_{s_N}^{\alpha_{N-1}}$
1605.05775_ EQ0008 (9)	004	■ 0.999 ● C +4	$\begin{aligned} & \Delta B^{\ell} \stackrel{\text { def }}{=} -\frac{\partial C}{\partial B^{\ell}} \\ & =\sum_{n=1}^{N_T} \sum_{\ell'}\left(\delta_{L_n}^{\ell'}-f^{\ell'}\left(\mathbf{x}_n\right)\right) \frac{\partial f^{\ell'}\left(\mathbf{x}_n\right)}{\partial B^{\ell}} \\ & =\sum_{n=1}^{N_T}\left(\delta_{L_n}^{\ell}-f^{\ell}\left(\mathbf{x}_n\right)\right) \tilde{\Phi}_n \end{aligned}$	$\begin{aligned} \Delta B^{\ell} &\stackrel{\text { def }}{=} -\frac{\partial C}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T} \sum_{\ell'}\left(\delta_{L_n}^{\ell'}-f^{\ell'}\left(\mathbf{x}_n\right)\right) \frac{\partial f^{\ell'}\left(\mathbf{x}_n\right)}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T}\left(\delta_{L_n}^{\ell}-f^{\ell}\left(\mathbf{x}_n\right)\right) \tilde{\Phi}_n \end{aligned}$	$\begin{aligned} \Delta B^{\ell} &\stackrel{\text { def }}{=} -\frac{\partial C}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T} \sum_{\ell'}\left(\delta_{L_n}^{\ell'}-f^{\ell'}\left(\mathbf{x}_n\right)\right) \frac{\partial f^{\ell'}\left(\mathbf{x}_n\right)}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T}\left(\delta_{L_n}^{\ell}-f^{\ell}\left(\mathbf{x}_n\right)\right) \tilde{\Phi}_n \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
1605.05775_ EQ0009 (11)	004	■ 0.999 ● W +1	$B_{s_j s_{j+1}}^{\alpha_j s_{j+1}} = \sum_{\alpha_j} U_{s_j \alpha_j'}^{\alpha_j-1} S_{\alpha_j}^{\alpha_j'} V_{s_{j+1}}^{\alpha_j \ell \alpha_{j+1}}$	$B_{s_j s_{j+1}}^{\alpha_j-1 \ell \alpha_{j+1}} = \sum_{\alpha_j' \alpha_j} U_{s_j \alpha_j'}^{\alpha_j-1} S_{\alpha_j}^{\alpha_j'} V_{s_{j+1}}^{\alpha_j \ell \alpha_{j+1}}$	$B_{s_j s_{j+1}}^{\alpha_j-1 \ell \alpha_{j+1}} = \sum_{\alpha_j' \alpha_j} U_{s_j \alpha_j'}^{\alpha_j-1} S_{\alpha_j}^{\alpha_j'} V_{s_{j+1}}^{\alpha_j \ell \alpha_{j+1}},$
1605.05775_ EQ0001 (1)	001	■ 1.000 ● N +0	$f(\mathbf{x})=W \cdot \Phi(\mathbf{x})$	$f(\mathbf{x}) = W \cdot \Phi(\mathbf{x}).$	$f(\mathbf{x}) = W \cdot \Phi(\mathbf{x})$
1605.05775_ EQ0002 (2)	002	■ 1.000 ● W +0	$\Phi^{s_1 s_2 \cdots s_N}(\mathbf{x}) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$	$\Phi^{s_1 s_2 \cdots s_N}(\mathbf{x}) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N).$	$\Phi^{s_1 s_2 \cdots s_N}(\mathbf{x}) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$
1605.05775_ EQ0003 (3)	002	■ 1.000 ● N +0	$\phi^{s_j}(x_j) = \left[\cos\left(\frac{\pi}{2}x_j\right), \sin\left(\frac{\pi}{2}x_j\right) \right]$	$\phi^{s_j}(x_j) = \left[\cos\left(\frac{\pi}{2}x_j\right), \sin\left(\frac{\pi}{2}x_j\right) \right]$	$\phi^{s_j}(x_j) = \left[\cos\left(\frac{\pi}{2}x_j\right), \sin\left(\frac{\pi}{2}x_j\right) \right]$
1605.05775_ EQ0004 (4)	002	■ 1.000 ● K +0	$f^{\ell}(\mathbf{x})=W^{\ell} \cdot \Phi(\mathbf{x})$	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$
1605.05775_ EQ0006 (6)	003	■ 1.000 ● N +1	$C=\frac{1}{2} \sum_{n=1}^{N_T} \sum_{\ell} (f^{\ell}(\mathbf{x}_n) - \delta_{L_n}^{\ell})^2$	$C = \frac{1}{2} \sum_{n=1}^{N_T} \sum_{\ell} (f^{\ell}(\mathbf{x}_n) - \delta_{L_n}^{\ell})^2$	$C = \frac{1}{2} \sum_{n=1}^{N_T} \sum_{\ell} (f^{\ell}(\mathbf{x}_n) - \delta_{L_n}^{\ell})^2$
1605.05775_ EQ0007 (7)	004	■ 1.000 ● W +1	$f^{\ell}(\mathbf{x}_n) = \sum_{\alpha_j-1 \alpha_{j+1}} \sum_{s_j s_{j+1}} B_{s_j s_{j+1}}^{\alpha_j-1 \ell \alpha_{j+1}} \left(\tilde{\Phi}_n \right)_{\alpha_j-1 \ell \alpha_{j+1}}^{s_j s_{j+1+1}}$	$f^{\ell}(\mathbf{x}_n) = \sum_{\alpha_j-1 \alpha_{j+1}} \sum_{s_j s_{j+1}} B_{s_j s_{j+1}}^{\alpha_j-1 \ell \alpha_{j+1}} \left(\tilde{\Phi}_n \right)_{\alpha_j-1 \ell \alpha_{j+1}}^{s_j s_{j+1+1}}$	$f^{\ell}(\mathbf{x}_n) = \sum_{\alpha_j-1 \alpha_{j+1}} \sum_{s_j s_{j+1}} B_{s_j s_{j+1}}^{\alpha_j-1 \ell \alpha_{j+1}} \left(\tilde{\Phi}_n \right)_{\alpha_j-1 \ell \alpha_{j+1}}^{s_j s_{j+1+1}}$
1605.05775_ EQ0010 (12)	005	■ 1.000 ● N +0	$\Phi\left(x_1,x_2\right)=\phi^{s_1}\left(x_1\right) \otimes \phi^{s_2}\left(x_2\right)$	$\Phi\left(x_1,x_2\right)=\phi^{s_1}\left(x_1\right) \otimes \phi^{s_2}\left(x_2\right)$	$\Phi\left(x_1,x_2\right)=\phi^{s_1}\left(x_1\right) \otimes \phi^{s_2}\left(x_2\right)$
1605.05775_ EQ0011 (13)	007	■ 1.000 ● N +0	$f(\mathbf{x})=\sum_{\{s\}} W_{s_1 s_2 \cdots s_N} \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$	$f(\mathbf{x}) = \sum_{\{s\}} W_{s_1 s_2 \cdots s_N} \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N).$	$f(\mathbf{x}) = \sum_{\{s\}} W_{s_1 s_2 \cdots s_N} \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N).$
1605.05775_ EQ0012 (14)	007	■ 1.000 ● N +0	$\phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$	$\phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$	$\phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$
1605.05775_ EQ0015 (18)	008	■ 1.000 ● N +0	$\sum_{\ell} \int_{\mathbf{x}} f^{\ell}(\mathbf{x}) ^2 d\mu(x) = 1$	$\sum_{\ell} \int_{\mathbf{x}} f^{\ell}(\mathbf{x}) ^2 d\mu(x) = 1$	$\sum_{\ell} \int_{\mathbf{x}} f^{\ell}(\mathbf{x}) ^2 d\mu(x) = 1$
1605.05775_ EQ0016 (19)	008	■ 1.000 ● W +0	$\sum_{\ell} \sum_{s_1, s_2, \dots, s_N} \bar{W}_{s_1 s_2 \cdots s_N}^{\ell} W_{s_1 s_2 \cdots s_N}^{\ell} = 1.$	$\sum_{\ell} \sum_{s_1, s_2, \dots, s_N} \bar{W}_{s_1 s_2 \cdots s_N}^{\ell} W_{s_1 s_2 \cdots s_N}^{\ell} = 1.$	$\sum_{\ell} \sum_{s_1, s_2, \dots, s_N} \bar{W}_{s_1 s_2 \cdots s_N}^{\ell} W_{s_1 s_2 \cdots s_N}^{\ell} = 1.$
1605.05775_ EQ0017 (20)	008	■ 1.000 ● N +1	$\int_x \bar{\phi}^s(x) \phi^{s'}(x) d\mu(x) = \delta_{ss'}$	$\int_x \bar{\phi}^s(x) \phi^{s'}(x) d\mu(x) = \delta_{ss'}$	$\int_x \bar{\phi}^s(x) \phi^{s'}(x) d\mu(x) = \delta_{ss'},$
1605.05775_ EQ0018 (21)	008	■ 1.000 ● N +0	$\sum_s \phi^s(x) ^2 = 1$	$\sum_s \phi^s(x) ^2 = 1$	$\sum_s \phi^s(x) ^2 = 1$
1605.05775_ EQ0019 (22)	008	■ 1.000 ● N +0	$\phi(x)=[\cos(\pi x), \sin(\pi x)]$	$\phi(x) = [\cos(\pi x), \sin(\pi x)].$	$\phi(x) = [\cos(\pi x), \sin(\pi x)].$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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1605.05775_ EQ0020 (23)	008	■ 1.000 ● N +0	$\phi(x)=\left[e^{i(3\pi/2)x}\cos\left(\frac{\pi}{2}x\right),e^{-i(3\pi/2)x}\sin\left(\frac{\pi}{2}x\right)\right]$.	$\phi(x)=\left[e^{i(3\pi/2)x}\cos\left(\frac{\pi}{2}x\right),e^{-i(3\pi/2)x}\sin\left(\frac{\pi}{2}x\right)\right]$.	$\phi(x)=\left[e^{i(3\pi/2)x}\cos\left(\frac{\pi}{2}x\right),e^{-i(3\pi/2)x}\sin\left(\frac{\pi}{2}x\right)\right]$.
1605.05775_ EQ0021 (24)	009	■ 1.000 ● N +1	$C=-\sum_{n=1}^{N_s}\log\left f^{L_n}(\mathbf{x}_n)\right ^2$	$C=-\sum_{n=1}^{N_s}\log\left f^{L_n}(\mathbf{x}_n)\right ^2$	$C=-\sum_{n=1}^{N_s}\log\left f^{L_n}(\mathbf{x}_n)\right ^2$
1605.05775_ EQ0022 (25)	009	■ 1.000 ● N +1	$D_{\mathrm{KL}}=\sum_{\ell}\int_{\mathbf{x}}p(\ell,\mathbf{x})\log\left(\frac{p(\ell,\mathbf{x})}{\tilde{p}(\ell,\mathbf{x})}\right)$	$D_{\mathrm{KL}}=\sum_{\ell}\int_{\mathbf{x}}p(\ell,\mathbf{x})\log\left(\frac{p(\ell,\mathbf{x})}{\tilde{p}(\ell,\mathbf{x})}\right)$	$D_{\mathrm{KL}}=\sum_{\ell}\int_{\mathbf{x}}p(\ell,\mathbf{x})\log\left(\frac{p(\ell,\mathbf{x})}{\tilde{p}(\ell,\mathbf{x})}\right)$
1605.05775_ EQ0023 (B1)	010	■ 1.000 ● N +0	$\phi^{s_j}(x_j)=\left[\cos\left(\frac{\pi}{2}x_j\right),\sin\left(\frac{\pi}{2}x_j\right)\right]$.	$\phi^{s_j}(x_j)=\left[\cos\left(\frac{\pi}{2}x_j\right),\sin\left(\frac{\pi}{2}x_j\right)\right]$.	$\phi^{s_j}(x_j)=\left[\cos\left(\frac{\pi}{2}x_j\right),\sin\left(\frac{\pi}{2}x_j\right)\right]$.
1605.05775_ EQ0024 (B2)	010	■ 1.000 ● N +0	$\left(\cos^2(\theta_j)+\sin^2(\theta_j)\right)^{d-1}=1$.	$\left(\cos^2(\theta_j)+\sin^2(\theta_j)\right)^{d-1}=1$.	$(\cos^2(\theta_j)+\sin^2(\theta_j))^{d-1}=1$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
1605.05775_F00001	001	—	$\{ \ }^{\{1,2\}}$	$1,2$
1605.05775_F00002	001	—	$\backslash\mathrm{Phi}(\backslash\mathrm{mathbf{x}})$	$\Phi(\mathbf{x})$
1605.05775_F00003	001	—	W	W
1605.05775_F00004	001	—	$\backslash\mathrm{Phi}$	Φ
1605.05775_F00005	002	—	d	d
1605.05775_F00006	002	—	N	N
1605.05775_F00007	002	—	$\backslash\mathrm{Phi}^{\{s_{\{1\}}\} \ s_{\{2\}} \ \cdots \ s_{\{N\}}\}}$	$\Phi^{s_1 s_2 \cdots s_N}$
1605.05775_F00008	002	—	$\backslash\mathrm{phi}^{\{s_{\{j\}}\}}\backslash\mathrm{left}(x_{\{j\}}\backslash\mathrm{right})$	$\phi^{s_j}(x_j)$
1605.05775_F00009	002	—	$x_{\{j\}}$	x_j
1605.05775_F00010	002	—	$s_{\{j\}}$	s_j
1605.05775_F00011	002	—	$d^{\{N\}}$	d^N
1605.05775_F00012	002	—	$j^{\backslash\mathrm{text}\ \mathrm{th}\ }}$	j^{th}
1605.05775_F00013	002	—	$x_{\{j\}} \ \backslash\mathrm{in}[0,1]$	$x_j \in [0,1]$
1605.05775_F00014	002	—	$\backslash\mathrm{phi}^{\{s_{\{j\}}\}}$	ϕ^{s_j}
1605.05775_F00015	002	—	$d=2$	$d=2$
1605.05775_F00016	002	—	$\backslash\mathrm{ell}$	ℓ
1605.05775_F00017	002	—	$\backslash\mathrm{mathbf{x}}$	$\mathbf{x}$
1605.05775_F00018	002	—	$\backslash\mathrm{left} f^{\backslash\mathrm{ell}}(\backslash\mathrm{mathbf{x}})\backslash\mathrm{right} $	$ f^{\ell}(\mathbf{x}) $
1605.05775_F00019	002	—	$W^{\backslash\mathrm{ell}}$	W^{ℓ}
1605.05775_F00020	002	—	$N+1$	$N+1$
1605.05775_F00021	003	—	$f^{\backslash\mathrm{ell}}(\backslash\mathrm{mathbf{x}})$	$f^{\ell}(\mathbf{x})$
1605.05775_F00022	003	—	$W_{\{s_{\{1\}}\} \ s_{\{2\}} \ \cdots \ s_{\{N\}}\}}^{\backslash\mathrm{ell}}$	$W_{s_1 s_2 \cdots s_N}^{\ell}$
1605.05775_F00023	003	—	$N_{\{L\}} \ \backslash\mathrm{cdot} \ d^{\{N\}}$	$N_L \cdot d^N$
1605.05775_F00024	003	—	$N_{\{L\}}$	N_L
1605.05775_F00025	003	—	$\backslash\mathrm{alpha}_{\{j\}}$	α_j
1605.05775_F00026	003	—	m	m
1605.05775_F00027	003	—	$\backslash\mathrm{left}\{\backslash\mathrm{alpha}_{\{j\}}\backslash\mathrm{right}\}$	$\{\alpha_j\}$
1605.05775_F00028	003	—	n	n
1605.05775_F00029	003	—	$N_{\{T\}}$	N_T
1605.05775_F00030	003	—	$L_{\{n\}}$	L_n
1605.05775_F00031	003	—	$\backslash\mathrm{delta}_{\{L_{\{n\}}\}}^{\backslash\mathrm{ell}}=1$	$\delta_{L_n}^{\ell}=1$
1605.05775_F00032	003	—	$f^{\backslash\mathrm{ell}}$	f^{ℓ}
1605.05775_F00033	003	—	$\backslash\mathrm{mathbf{x}}_{\{n\}}$	$\mathbf{x}_n$
1605.05775_F00034	003	—	j	j
1605.05775_F00035	003	—	$j+1$	$j+1$
1605.05775_F00036	003	—	$A_{\{s_{\{j\}}\}}^{\backslash\mathrm{ell}}$	$A_{s_j}^{\ell}$
1605.05775_F00037	003	—	$A_{\{s_{\{j+1\}}\}}$	$A_{s_{j+1}}$
1605.05775_F00038	004	—	$B^{\backslash\mathrm{ell}}$	B^{ℓ}
1605.05775_F00039	004	—	$B_{\{s_{\{j\}}\} \ s_{\{j+1\}}\}}^{\backslash\mathrm{alpha}_{\{j-1\}} \ \mathrm{ell} \ \backslash\mathrm{alpha}_{\{j+1\}}\}}$	$B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}}$
1605.05775_F00040	004	—	C	C
1605.05775_F00041	004	—	$\backslash\mathrm{tilde}\{\backslash\mathrm{Phi}\}_{\{n\}}$	$\tilde{\Phi}_n$
1605.05775_F00042	004	—	$B^{\backslash\mathrm{ell}}+\backslash\mathrm{alpha} \ \Delta B^{\backslash\mathrm{ell}}$	$B^{\ell}+\alpha \Delta B^{\ell}$
1605.05775_F00043	004	—	$\backslash\mathrm{alpha}$	α
1605.05775_F00044	004	—	$j+1)$	$j+1)$
1605.05775_F00045	004	—	$\backslash\mathrm{left}(\backslash\mathrm{alpha}_{\{j-1\}}, \ s_{\{j\}}\backslash\mathrm{right})$	(α_{j-1}, s_j)
1605.05775_F00046	004	—	$\backslash\mathrm{operatorname{dex}}\backslash\mathrm{left}(\backslash\mathrm{ell}, \ \backslash\mathrm{alpha}_{\{j+1\}}, \ s_{\{j+1\}}\backslash\mathrm{right})$	$\mathrm{dex}(\ell, \alpha_{j+1}, s_{j+1})$
1605.05775_F00047	004	—	$A_{\{s_{\{j\}}\}}^{\backslash\mathrm{prime}}=U_{\{s_{\{j\}}\}}$	$A'_{s_j}=U_{s_j}$
1605.05775_F00048	004	—	$A_{\{s_{\{j+1\}}\}}^{\backslash\mathrm{prime} \ \backslash\mathrm{ell}}=S \ V_{\{s_{\{j+1\}}\}}^{\backslash\mathrm{ell}}$	$A_{s_{j+1}}^{\ell}=S V_{s_{j+1}}^{\ell}$
1605.05775_F00049	004	—	S	S
1605.05775_F00050	004	—	U	U
1605.05775_F00051	004	—	$V^{\backslash\mathrm{dagger}}$	$V^{\dagger}$
1605.05775_F00052	004	—	$A_{\{s_{\{j\}}\}}^{\backslash\mathrm{prime}}$	A'_{s_j}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

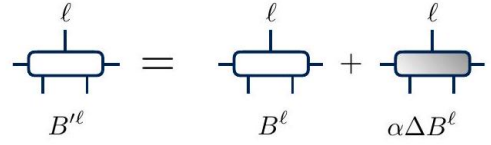
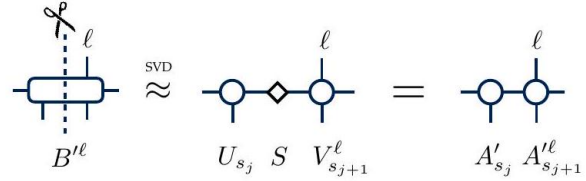
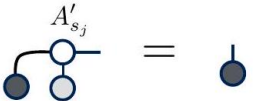
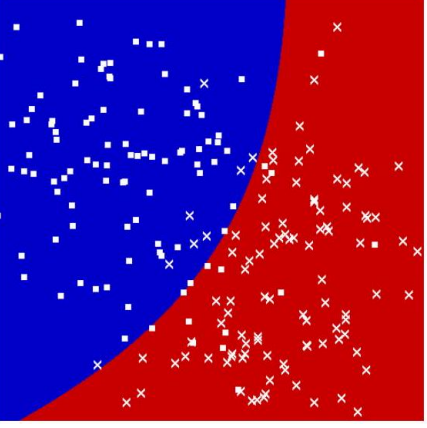
Identifier	Page	Conf.	LaTeX source	Rendered
1605.05775_F00053	005	—	$B^{\prime \ell}$	B^{ℓ}
1605.05775_F00054	005	—	$d^3 m^3 N N_L N_T$	$d^3 m^3 N N_L N_T$
1605.05775_F00055	005	—	14×14	14×14
1605.05775_F00056	005	—	$s_{\{1\}}, s_{\{2\}}, \ldots, s_{\{N\}}$	$s_1, s_2, \ldots, s_N$
1605.05775_F00057	005	—	$m=10$	$m = 10$
1605.05775_F00058	005	—	$m=20$	$m = 20$
1605.05775_F00059	005	—	$m=120$	$m = 120$
1605.05775_F00060	005	—	$(N=2)$	$(N = 2)$
1605.05775_F00061	005	—	$P_{\{A\}}\!\left(x_{\{1\}}, x_{\{2\}}\right)$	$P_A(x_1, x_2)$
1605.05775_F00062	005	—	$P_{\{B\}}\!\left(x_{\{1\}}, x_{\{2\}}\right)$	$P_B(x_1, x_2)$
1605.05775_F00063	005	—	A	A
1605.05775_F00064	005	—	B	B
1605.05775_F00065	005	—	$x_{\{1\}} \in [0,1]$	$x_1 \in [0,1]$
1605.05775_F00066	005	—	$x_{\{2\}} \in [0,1]$	$x_2 \in [0,1]$
1605.05775_F00067	005	—	$\left(x_{\{1\}}, x_{\{2\}}\right)$	(x_1, x_2)
1605.05775_F00068	005	—	$W_{s_{\{1\}} s_{\{2\}}}^{\ell}$	$W_{s_1 s_2}^{\ell}$
1605.05775_F00069	005	—	$\ell \in \{A, B\}$	$\ell \in \{A, B\}$
1605.05775_F00070	005	—	$s_{\{1\}}$	s_1
1605.05775_F00071	005	—	$s_{\{2\}}$	s_2
1605.05775_F00072	005	—	$d>2$	$d > 2$
1605.05775_F00073	005	—	$P_{\{A\}}$	P_A
1605.05775_F00074	005	—	$P_{\{B\}}$	P_B
1605.05775_F00075	006	—	$P_{\{A\}}\!\left(x_{\{1\}}, x_{\{2\}}\right)=P_{\{B\}}\!\left(x_{\{1\}}, x_{\{2\}}\right)$	$P_A(x_1, x_2) = P_B(x_1, x_2)$
1605.05775_F00076	006	—	$d=3$	$d = 3$
1605.05775_F00077	006	—	$d=6$	$d = 6$
1605.05775_F00078	006	—	$d=10$	$d = 10$
1605.05775_F00079	006	—	$B)$	$B)$
1605.05775_F00080	007	—	$f^{\ell}(\mathbf{x})(\mathbf{x})=W^{\ell} \cdot \Phi(\mathbf{x})$	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$
1605.05775_F00081	007	—	$f(\mathbf{x})$	$f(\mathbf{x})$
1605.05775_F00082	007	—	$\left\{\phi^s(x)\right\}, s=1,2,\ldots,d$	$\{\phi^s(x)\}, s = 1, 2, \ldots, d$
1605.05775_F00083	007	—	$x \in [0,1]$	$x \in [0,1]$
1605.05775_F00084	007	—	$\mathbf{x} \in [0,1]^{\times N}$	$\mathbf{x} \in [0,1]^{\times N}$
1605.05775_F00085	007	—	$\left\{\phi^s(x)\right\}$	$\{\phi^s(x)\}$
1605.05775_F00086	007	—	$\phi(x)$	$\phi(x)$
1605.05775_F00087	007	—	$\hat{\mathbf{x}}$	$\hat{\mathbf{x}}$
1605.05775_F00088	007	—	$\hat{x}_j=\{0,1\}$	$\hat{x}_j = \{0,1\}$
1605.05775_F00089	007	—	$f(\hat{\mathbf{x}})$	$f(\hat{\mathbf{x}})$
1605.05775_F00090	007	—	$W_{s_{\{1\}} s_{\{2\}} \ldots s_{\{N\}}}$	$W_{s_1 s_2 \ldots s_N}$
1605.05775_F00091	007	—	$x_{\{j\}} \in [0,1], f(\mathbf{x})$	$x_j \in [0,1], f(\mathbf{x})$
1605.05775_F00092	007	—	$a \cos \left(\pi / 2 x_{\{j\}}\right)+b \sin \left(\pi / 2 x_{\{j\}}\right)$	$a \cos (\pi / 2 x_j)+b \sin (\pi / 2 x_j)$
1605.05775_F00093	007	—	a	a
1605.05775_F00094	007	—	b	b
1605.05775_F00095	007	—	$C^{\alpha_i \ell \alpha_{i+1}}$	$C^{\alpha_i \ell \alpha_{i+1}}$
1605.05775_F00096	007	—	i	i
1605.05775_F00097	007	—	$j \leq i$	$j \leq i$
1605.05775_F00098	007	—	$j \geq i$	$j \geq i$
1605.05775_F00099	007	—	V	V
1605.05775_F00100	007	—	$U_{\alpha_{j-1} s_{\{j\}}}^{\alpha_j}$	$U_{\alpha_{j-1} s_j}^{\alpha_j}$
1605.05775_F00101	007	—	$V^{\alpha_{j-1}}_{s_{\{j\}} \alpha_j}$	$V^{\alpha_{j-1}}_{s_j \alpha_j}$
1605.05775_F00102	007	—	$U^{\dagger} U=I$	$U^{\dagger} U = I$
1605.05775_F00103	007	—	$V V^{\dagger}=I$	$V V^{\dagger} = I$
1605.05775_F00104	007	—	I	I
1605.05775_F00105	007	—	m^2	m^2
1605.05775_F00106	007	—	$\tilde{\Phi}(\mathbf{x})$	$\tilde{\Phi}(\mathbf{x})$
1605.05775_F00107	007	—	C^{ℓ}	C^{ℓ}
1605.05775_F00108	007	—	α_i	α_i
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

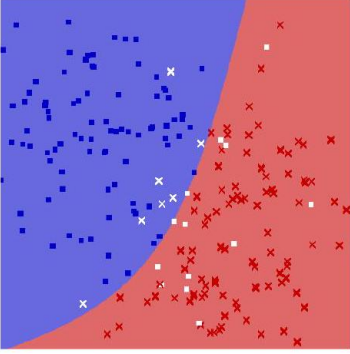
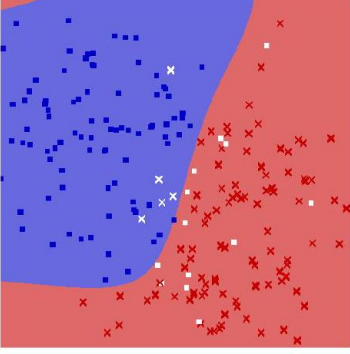
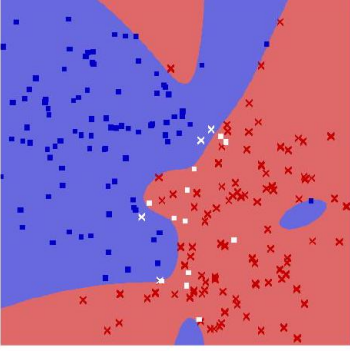
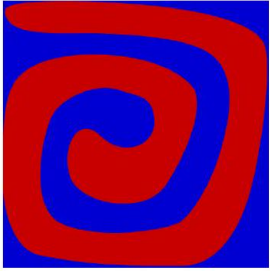
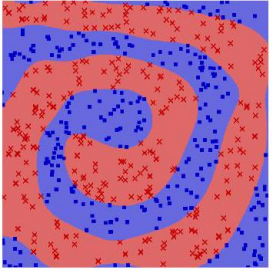
Identifier	Page	Conf.	LaTeX source	Rendered
1605.05775_F00109	007	—	$\backslash\alpha_{i+1}$	α_{i+1}
1605.05775_F00110	008	—	$U_{\backslash\alpha_{\{j\}}\ s_{\{j\}}}^{\alpha_{\{j\}}+1}$	$U_{\alpha_j s_j}^{\alpha_j+1}$
1605.05775_F00111	008	—	$\backslash\alpha_{j+1}$	α_{j+1}
1605.05775_F00112	008	—	$\left f^{\ell}(\mathbf{x})\right ^2$	$\left f^{\ell}(\mathbf{x})\right ^2$
1605.05775_F00113	008	—	$d\ \mu(x)$	$d\mu(x)$
1605.05775_F00114	008	—	$\phi^s(x)$	$\phi^s(x)$
1605.05775_F00115	008	—	ϕ^s	ϕ^s
1605.05775_F00116	008	—	$d\ \mu(x)=2\ d\ x$	$d\mu(x)=2dx$
1605.05775_F00117	008	—	$N_{\{s\}}$	N_s
1605.05775_F00118	008	—	$\ell=A$	$\ell=A$
1605.05775_F00119	008	—	$\ell=B$	$\ell=B$
1605.05775_F00120	008	—	$P_{\{A\}}=\int\limits_{\mathbf{x}}\left f^A(\mathbf{x})\right ^2=\sum_{s_1\ s_2}\left W_{s_1\ s_2}^A\right ^2$	$P_A=\int_{\mathbf{x}}\left f^A(\mathbf{x})\right ^2=\sum_{s_1\ s_2}\left W_{s_1\ s_2}^A\right ^2$
1605.05775_F00121	008	—	$P_{\{B\}}=1-P_{\{A\}}$	$P_B=1-P_A$
1605.05775_F00122	009	—	$\sigma\ /\ \sqrt{N_{\{s\}}}$	$\sigma/\sqrt{N_s}$
1605.05775_F00123	009	—	$\mathbf{x}=\left(x_1,\ x_2\right)$	$\mathbf{x}=(x_1,x_2)$
1605.05775_F00124	009	—	$p(\mathbf{x}\mid\ell)=\left f^{\ell}(\mathbf{x})\right ^2\ /\ P_{\ell}$	$p(\mathbf{x}\mid\ell)=\left f^{\ell}(\mathbf{x})\right ^2/P_{\ell}$
1605.05775_F00125	009	—	$\widetilde{W}$	$\widetilde{W}$
1605.05775_F00126	009	—	$p(\ell,\mathbf{x})=\left f^{\ell}(\mathbf{x})\right ^2=\left W^{\ell}\cdot\Phi(\mathbf{x})\right ^2$	$p(\ell,\mathbf{x})=\left f^{\ell}(\mathbf{x})\right ^2=\left W^{\ell}\cdot\Phi(\mathbf{x})\right ^2$
1605.05775_F00127	009	—	$\widetilde{p}(\ell,\mathbf{x})$	$\widetilde{p}(\ell,\mathbf{x})$
1605.05775_F00128	009	—	$D_{\{\text{KL}\}}$	D_{KL}
1605.05775_F00129	009	—	σ	σ
1605.05775_F00130	009	—	$L_{\{1\}}$	L_1
1605.05775_F00131	010	—	$\phi^{s_j}(x_j),\ s_j=1,2,\ \ldots,\ d$	$\phi^{s_j}(x_j),\ s_j=1,2,\ldots,d$
1605.05775_F00132	010	—	$\theta_j=\frac{\pi}{2}x_j$	$\theta_j=\frac{\pi}{2}x_j$
1605.05775_F00133	010	—	$\left(\cos^2\left(\theta_j\right)+\sin^2\left(\theta_j\right)\right)=1$	$\left(\cos^2\left(\theta_j\right)+\sin^2\left(\theta_j\right)\right)=1$
1605.05775_F00134	010	—	$\binom{n}{k}=n!/(k!(n-k)!)$	$\binom{n}{k}=n!/(k!(n-k)!)$
1605.05775_F00135	010	—	$\sum_{s_j}\left \phi^{s_j}\right ^2=1$	$\sum_{s_j}\left \phi^{s_j}\right ^2=1$
1605.05775_F00136	010	—	$W^{\ell}\cdot\Phi(\mathbf{x})$	$W^{\ell}\cdot\Phi(\mathbf{x})$
1605.05775_F00137	010	—	$\cos\left(\frac{\pi}{2}x_j\right),\cos\left(\frac{3\pi}{2}x_j\right),\ldots$	$\cos\left(\frac{\pi}{2}x_j\right),\cos\left(\frac{3\pi}{2}x_j\right),\ldots$
1605.05775_F00138	010	—	$\cos\left(\frac{(d-1)\pi}{2}x_j\right)$	$\cos\left(\frac{(d-1)\pi}{2}x_j\right)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
1605.05775_PIC_0001	009		tex.zip (local)
1605.05775_DIA_0001	001		tex.zip (local)
1605.05775_DIA_0002	002	$\Phi = \begin{matrix} s_1 & s_2 & s_3 & s_4 & s_5 & s_6 \\ \circ & \circ & \circ & \circ & \circ & \circ \\ \phi^{s_1} & \phi^{s_2} & \phi^{s_3} & \phi^{s_4} & \phi^{s_5} & \phi^{s_6} \end{matrix}$	tex.zip (local)
1605.05775_DIA_0003	002		tex.zip (local)
1605.05775_DIA_0004	003	$\begin{matrix} \ell \\ \text{---} \\ \circ \quad \circ \quad \circ \quad \circ \quad \circ \quad \circ \end{matrix} W^\ell = \begin{matrix} \ell \\ \text{---} \\ \square \end{matrix} f^\ell(\mathbf{x})$	tex.zip (local)
1605.05775_DIA_0005	003		tex.zip (local)
1605.05775_DIA_0006	004	(a) $\begin{matrix} \ell \\ \text{---} \\ \circ \quad \circ \quad \circ \quad \circ \quad \circ \quad \circ \end{matrix} = \begin{matrix} \ell \\ \text{---} \\ \circ \quad \circ \quad \text{---} \quad \circ \quad \circ \end{matrix}$ (b) $\begin{matrix} \circ \quad \circ \quad \circ \quad \circ \quad \circ \quad \circ \end{matrix} \Phi_n = \begin{matrix} \bullet \quad \circ \quad \circ \quad \bullet \end{matrix} \tilde{\Phi}_n$ (c) $\begin{matrix} \ell \\ \text{---} \\ \bullet \quad \circ \quad \circ \quad \bullet \end{matrix} B^\ell = \begin{matrix} \ell \\ \text{---} \\ \square \end{matrix} f^\ell(\mathbf{x}_n)$ (d) $\Delta B^\ell = \sum_n \begin{matrix} \ell \\ \text{---} \\ \bullet \quad \circ \quad \circ \quad \bullet \end{matrix} (\delta_{L_n}^\ell - f^\ell(\mathbf{x}_n))$	tex.zip (local)

Identifier	Page	Image	Source
1605.05775_DIA_0007	004	(a)  (b)  (c) 	tex.zip (local)
1605.05775_DIA_0008	005		tex.zip (local)
1605.05775_DIA_0009	006		tex.zip (local)

Identifier	Page	Image	Source
1605.05775_DIA_0010	006	(a)  (b)  (c) 	tex.zip (local)
1605.05775_DIA_0011	007	(a)  (b) 	tex.zip (local)
1605.05775_DIA_0012	008	(a) $W_{s_1 \dots s_6}^\ell = \begin{array}{c} U_{s_1} \ U_{s_2} \ U_{s_3} \ \ell \\ \circ - \circ - \circ - \diamond - \circ - \circ - \circ \\ \quad \quad \quad C \end{array}$ (b) $\begin{array}{c} U_{s_j} \\ \circ \\ \circ \\ U_{s_j}^\dagger \end{array} = \text{C} \quad \begin{array}{c} V_{s_j} \\ \circ \\ \circ \\ V_{s_j}^\dagger \end{array} = \text{C}$ (c) $\begin{array}{c} \circ - \circ - \circ \\ \circ \\ \circ \end{array} \begin{array}{c} \circ - \circ - \circ \\ \circ \\ \circ \end{array} \Phi(x) = \begin{array}{c} \parallel \\ \circ \end{array} \tilde{\Phi}(x)$	tex.zip (local)

Identifier	Page	Image	Source
1605.05775_DIA_0013	010		tex.zip (local)
1605.05775_DIA_0014	010	(a) $\sum_j M_{ij} v_j = w_i$ (b) $\sum_{kl} A_{ijkl} B_{klm} = C_{ijm}$	tex.zip (local)